Power

Exam code: 0625 & 0972

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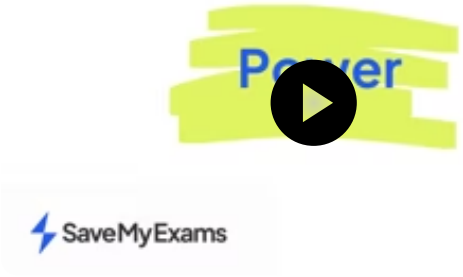

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Power

- Power is defined as:

Work done per unit time

- Since work done is equal to energy transferred, **power** is also defined as:

Energy transferred per unit time

- Machines, such as car engines, transfer energy from one energy store to another constantly over a period of **time**
  - The **rate** of this energy transfer, or the rate of work done, is **power**

- **Time** is an important consideration when it comes to **power**
- Two cars transfer the **same amount of energy**, or do the **same amount of work** to accelerate over a distance
- If one car has **more power**, it will transfer that energy, or do that work, in a **shorter amount of time**

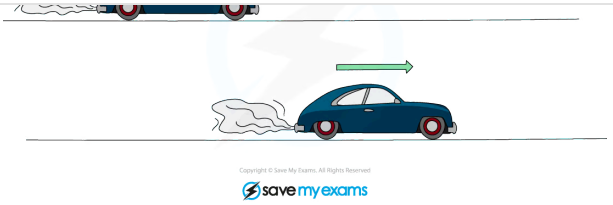
Two cars with different power ratings doing the same amount of work

Help me understand

Stuck on something?

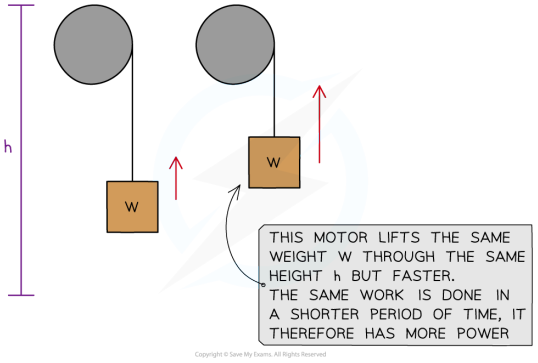
Ask questions and get help with exactly what you're revising.

Got it



Two cars accelerate to the same final speed, but the one with the most power will reach that speed sooner

- Two electric motors:
  - lift the same weight
  - by the same height
  - but one motor lifts it **faster** than the other
- The motor that lifts the weight faster has more **power**



Two motors with different powers

Power ratings

- **Power ratings** are given to appliances to show the amount of energy transferred per unit time
- Common power ratings are shown in the table below:

Power ratings table

Appliance	Power rating
A torch	1 W
An electric light bulb	100 W
An electric oven	10 000 W = 10 kW
A train	1 000 000 W = 1 MW
Saturn V space rocket	100 MW
Large power station	10 000 MW

A star like the Sun	100 000 000 000 000 000 000 MW

- 1 kW = 1000 W (1 kilowatt = 1000 watts)
- 1 MW = 1 000 000 W (1 megawatt = 1 million watts)

Calculating Power

The power equations

- There are two equivalent forms of the power equation
- Power can be expressed in terms of work done W:

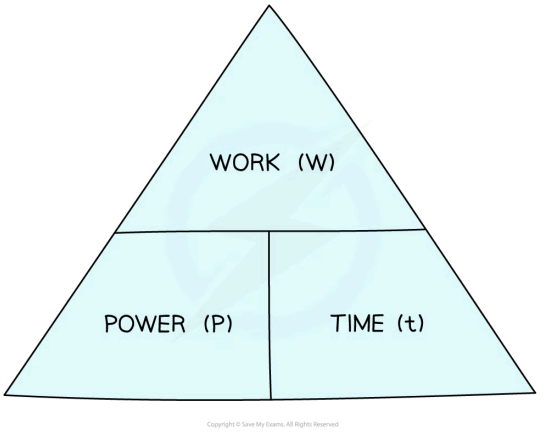
$$P = \frac{W}{t}$$

- Or, power can be expressed in terms of energy transferred ΔE:

$$P = \frac{\Delta E}{t}$$

- Where:
  - P = power measured in watts (W)
  - W = work done, measured in newton metres (N m)
  - t = time measured in seconds (s)
  - ΔE = energy transferred, measured in joules (J)
- Note that these two equations may be written slightly differently, but they represent the same thing
  - A transfer of energy (work done) over time

Power equation triangle



To use a formula triangle, simply cover up the quantity you wish to calculate and the structure of the equation is revealed

- Using an equation triangle is covered in more depth in the revision note Speed & velocity

Worked Example

Calculate the energy transferred when an iron with a power rating of 2000 W is used for 5 minutes.

Answer:

Step 1: List the known values

- Power, $P = 2000\ \text{W}$
- Time, $t = 5\ \text{minutes}$

Step 2: Convert minutes into seconds

- $1\ \text{min} = 60\ \text{s}$

$$5\ \text{min} \times 60\ \text{s} = 300\ \text{s}$$

Step 3: Write down the power equation in terms of energy

$$P = \frac{\Delta E}{t}$$

Step 4: Rearrange to make energy transferred, *ΔE, the subject*

$$\Delta E = Pt$$

Step 5: Substitute in the known values

$$\Delta E = 2000 \times 300$$

$$\Delta E = 600\ 000\ \text{J}$$

Examiner Tips and Tricks

Think of power as "energy per second". Thinking of it this way will help you to remember the relationship between power and energy.

In your IGCSE exam, you will be expected to use both equations and to be able to rearrange them. You may be required to calculate the energy transferred in a previous question part, so always check back through the question if you seem to be missing a value!

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